

Section 1: Hypertension Classification and Screening

Hypertension is a major preventable risk factor for cardiovascular disease, stroke, heart failure, and chronic kidney disease.

Blood pressure categories in adults: Normal: Systolic < 120 mmHg and Diastolic < 80 mmHg. Elevated BP: Systolic 120-129 mmHg and Diastolic < 80 mmHg. Stage 1 Hypertension: Systolic 130-139 mmHg or Diastolic 80-89 mmHg. Stage 2 Hypertension: Systolic $\geq$ 140 mmHg or Diastolic $\geq$ 90 mmHg. Diagnosis requires at least two separate seated office measurements on two separate occasions.

Section 2: Pharmacological Treatment of Hypertension

First-line antihypertensive drug classes include: Angiotensin-Converting Enzyme (ACE) Inhibitors (e.g., Lisinopril, Enalapril), Angiotensin II Receptor Blockers (ARBs, e.g., Losartan, Valsartan), Dihydropyridine Calcium Channel Blockers (CCBs, e.g., Amlodipine), and Thiazide/Thiazide-like diuretics (e.g., Chlorthalidone, Hydrochlorothiazide).

Monotherapy vs Combination: For Stage 2 Hypertension with BP $\geq$ 20/10 mmHg over goal, initiation with two first-line agents of different classes is recommended.

Target Blood Pressure: The primary therapeutic target for adults with confirmed hypertension and known CVD or 10-year ASCVD risk $\geq$ 10% is $<$ 130/80 mmHg.

Section 3: Guideline Variations and Target Ambiguities

Contrasting Guidelines and Target Blood Pressure Thresholds: While recent intensive treatment trials endorse a strict target of < 130/80 mmHg for high-risk cohorts, certain traditional guidelines (such as JNC-8 legacy protocols) recommend a blood pressure threshold of < 140/90 mmHg for general adult populations under age 60, and < 150/90 mmHg for individuals aged 60 and older without diabetes or CKD.

Clinical Resolution of Conflicts: Practitioners must evaluate individualized cardiovascular risk, orthostatic risk, and renal tolerance when setting target thresholds between 130/80 mmHg and 140/90 mmHg.